

Math 540

Homework #3

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§18 Homework Problems

Exercise 3. Let X and X' denote a single set in the two topologies T and T' , respectively. Let $i : X' \rightarrow X$ be the identity function.

- (a) Show that i is continuous if and only if T' is finer than T .
- (b) Show that i is a homeomorphism if and only if $T' = T$.

Proof. (a) Suppose i is continuous. Then for all $U \in T$, $U = i^{-1}(U) \in T'$, hence $T \subseteq T'$. Conversely, let $T \subseteq T'$. Let $U \in T$. Then $i^{-1}(U) = U \in T \subseteq T'$. Hence i is continuous.

(b) Suppose i is a homeomorphism. Then i and i^{-1} are continuous. Hence by part (a) $T \subseteq T'$ and $T' \subseteq T$. Thus $T = T'$. Conversely, suppose $T = T'$. Then $T \subseteq T'$ and $T' \subseteq T$. By part (a) we have that i is continuous and i^{-1} is continuous. Thus i is a homeomorphism. $\square$

Exercise 10. Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be continuous functions. Let us define a map $f \times g : A \times C \rightarrow B \times D$ by the equation $(f \times g)((a, c)) = (f(a), g(c))$. Show that $f \times g$ is continuous.

Proof. Note that the topology on $B \times D$ is generated by the basis $\mathcal{B} = \{U \times V \mid U \subseteq B \text{ is open, } V \subseteq D \text{ is open}\}$. Hence it suffices to prove that the preimage of a basis element is open in $A \times C$. Let $U \times V \in \mathcal{B}$. Then $(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V)$. Since $f^{-1}(U)$ is open in A and $g^{-1}(V)$ is open in C , then $f^{-1}(U) \times g^{-1}(V)$ is open in $A \times C$. Thus $f \times g$ is continuous. $\square$

Exercise 13. Let $A \subset X$; let $f : A \rightarrow Y$ be continuous; let Y be Hausdorff. Show that if f may be extended to a continuous function $g : \overline{A} \rightarrow Y$, then g is uniquely determined by f .

Proof. Suppose towards contradiction there exists continuous extensions $g, h : \overline{A} \rightarrow Y$ of f and there exists $x \in \overline{A}$ such that $g(x) \neq h(x)$. Since Y is Hausdorff, there exists open sets U and V such that $g(x) \in U$, $h(x) \in V$, and $U \cap V = \emptyset$. It follows that $x \in g^{-1}(U) \cap h^{-1}(V)$, which is an open set. But since $x \in \overline{A}$, it must be the case that $g^{-1}(U) \cap h^{-1}(V) \cap A \neq \emptyset$; i.e., there exists some $x' \in g^{-1}(U) \cap h^{-1}(V) \cap A$. Hence $g(x') \in U$, $h(x') \in V$, and $x' \in A$. But since g and h are extensions of f , we have that $g(x') = h(x')$. Hence $U \cap V \neq \emptyset$, which contradicts Y being Hausdorff. Thus any continuous extension of f is unique. $\square$

§20 Homework Problems

Exercise 1(a). In $\mathbb{R}^n$, define:

$$d'(x, y) = |x_1 - y_1| + \dots + |x_n - y_n|.$$

Show that d' is a metric that induces the usual topology of $\mathbb{R}^n$. Sketch the basis elements under d' when $n = 2$.

Proof. We will first show that d' is a metric. Note that:

$$\begin{aligned} d'(x, y) &= |x_1 - y_1| + \dots + |x_n - y_n| \\ &= |y_1 - x_1| + \dots + |y_n - x_n| \\ &= d'(y, x). \end{aligned}$$

Similarly, if $d'(x, y) = 0$, then $|x_1 - y_1| + \dots + |x_n - y_n| = 0$. Hence $|x_i - y_i| = 0$ for each $i = 1, 2, \dots, n$. Thus $x_i = y_i$ for all $i = 1, 2, \dots, n$. Hence $x = y$. Lastly, observe that:

$$\begin{aligned} d'(x, y) &= |x_1 - y_1| + \dots + |x_n - y_n| \\ &\leq |x_1 - z_1| + |z_1 - y_1| + \dots + |x_n - z_n| + |z_n - y_n| \\ &= (|x_1 - z_1| + \dots + |x_n - z_n|) + (|z_1 - y_1| + \dots + |z_n - y_n|) \\ &= d'(x, z) + d'(z, y). \end{aligned}$$

Thus d' is a metric on $\mathbb{R}^n$.

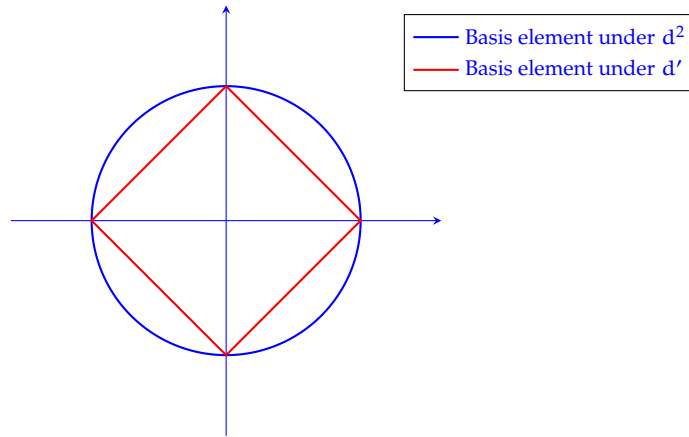
We will now show that d' induces the standard topology on $\mathbb{R}^n$. Note that the standard topology on $\mathbb{R}^n$ is induced by the metric $d^2(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^2\right)^{\frac{1}{2}}$. Observe that:

$$\begin{aligned} d^2(x, y) &= \left(\sum_{i=1}^n |x_i - y_i|^2\right)^{\frac{1}{2}} \\ &\leq \sum_{i=1}^n (|x_i - y_i|^2)^{\frac{1}{2}} \\ &= \sum_{i=1}^n |x_i - y_i| \\ &= d'(x, y). \end{aligned}$$

Let $x \in \mathbb{R}^n$ and $\epsilon > 0$ be arbitrary. If $y \in B_{d'}(x, \epsilon)$, then $d(x, y) < \epsilon$. Hence $d^2(x, y) < \epsilon$, which means $y \in B_{d^2}(x, \epsilon)$. Thus $B_{d'}(x, \epsilon) \subseteq B_{d^2}(x, \epsilon)$. Now observe that:

$$\begin{aligned} d'(x, y) &= \sum_{i=1}^n |x_i - y_i| \\ &\leq n \cdot \max_{i=1}^n |x_i - y_i| \\ &= n \cdot \left(\max_{i=1}^n |x_i - y_i|^2 \right)^{\frac{1}{2}} \\ &\leq n \cdot \left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{\frac{1}{2}} \\ &= n \cdot d^2(x, y). \end{aligned}$$

Let $x \in \mathbb{R}^n$ and $\epsilon > 0$ be arbitrary. If $y \in B_{d^2}(x, \frac{\epsilon}{n})$, then $n \cdot d^2(x, y) < \epsilon$. Hence $d'(x, y) < \epsilon$, which means $y \in B_{d'}(x, \epsilon)$. Thus $B_{d^2}(x, \frac{\epsilon}{n}) \subseteq B_{d'}(x, \epsilon)$. Therefore the topology induced by d' is the same by the topology induced by d^2 , which is the standard topology on $\mathbb{R}^n$.



□

Exercise 3. Let X be a metric space with metric d .

- (a) Show that $d : X \times X \rightarrow \mathbb{R}$ is continuous.
- (b) Let X' denote a space having the same underlying set as X . Show that if $d : X' \times X' \rightarrow \mathbb{R}$ is continuous, then the topology of X' is finer than the topology of X .

Proof. (a) Equip $X \times X$ with the product metric:

$$D((x_1, y_1), (x_2, y_2)) = d(x_1, y_1) + d(x_2, y_2).$$

Let $(x, y), (x', y') \in X \times X$. By the triangle inequality:

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y).$$

Hence $d(x, y) - d(x', y') \leq d(x, x') + d(y, y')$. Applying the triangle inequality similarly to $d(x', y')$ gives the inequality $d(x', y') - d(x, y) \leq d(x, x') + d(y, y')$. Hence:

$$\begin{aligned} |d(x, y) - d(x', y')| &\leq d(x, x') + d(y, y') \\ &= D((x, y), (x', y')). \end{aligned}$$

Thus d is Lipschitz, establishing d as continuous. $\square$

§21 Homework Problems

Exercise 1. Let $A \subset X$. If d is a metric for the topology of X , show that $d|_{A \times A}$ is a metric for the subspace topology on A .

Proof. Clearly the restriction of d onto $A \times A$ is a metric. Let $a \in A \cap U$, where $U \subseteq X$ is open. Then there exists some $r > 0$ such that $B_d(a, r) \subseteq U$. Hence $A \cap B_d(a, r) \subseteq A \cap U$. But $A \cap B_d(a, r) = B_{d|_{A \times A}}(a, r)$, hence $B_{d|_{A \times A}}(a, r) \subseteq A \cap U$. Thus $d|_{A \times A}$ is a metric for the subspace topology on A . $\square$

Exercise 2. Let X and Y be metric spaces with metrics d_X and d_Y , respectively. Let $f : X \rightarrow Y$ have the property that for every pair of points x_1, x_2 of X :

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

Show that f is an imbedding. (The above condition is called an isometry)

Proof. We must first show that f is continuous. Let $x_0 \in X$ and $\epsilon > 0$. Pick $\delta = \epsilon$. Then for any $x \in X$, $d_X(x, x_0) < \delta$ implies $d_Y(f(x), f(x_0)) = d_X(x, x_0) < \delta = \epsilon$.

Next, we must show that f is injective. Suppose $f(x) = f(x')$. Then $0 = d_Y(f(x), f(x')) = d_X(x, x')$. Hence $x = x'$, establishing that f is injective.

Since $f : X \rightarrow Y$ is continuous, $g : X \rightarrow f(X)$ is also continuous. In particular, since f is injective, g is bijective. We will now show that g^{-1} is an isometry. Observe that:

$$\begin{aligned} d_X(g^{-1}(y_0), g^{-1}(y)) &= d_X(g^{-1}(g(x_0)), g^{-1}(g(x))) \\ &= d_X(x_0, x) \\ &= d_Y(g(x_0), g(x)) \\ &= D_Y(y_0, y). \end{aligned}$$

Since g^{-1} is an isometry, it is continuous. Thus $g : X \rightarrow f(X)$ is a homeomorphism, establishing f as an imbedding. $\square$

Exercise 10. Using the closed set formulation of continuity, show that the following are closed subsets of $\mathbb{R}^2$:

$$\begin{aligned} A &= \{(x, y) \mid xy = 1\}, \\ S^1 &= \{(x, y) \mid x^2 + y^2 = 1\}, \\ B^2 &= \{(x, y) \mid x^2 + y^2 \leq 1\}. \end{aligned}$$

Proof. Define $f_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $(x, y) \mapsto xy$. This map is continuous, and:

$$\begin{aligned} f_1^{-1}(\{1\}) &= \{(x, y) \mid f((x, y)) \in \{1\}\} \\ &= \{(x, y) \mid xy = 1\} \\ &= A. \end{aligned}$$

Since $\{1\}$ is closed in $\mathbb{R}$, A is closed in $\mathbb{R}^2$.

Define $f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $(x, y) \mapsto x^2 + y^2$. This map is continuous, and:

$$\begin{aligned} f_2^{-1}(\{1\}) &= \{(x, y) \mid f((x, y)) \in \{1\}\} \\ &= \{(x, y) \mid x^2 + y^2 = 1\} \\ &= S^1. \end{aligned}$$

Since $\{1\}$ is closed in $\mathbb{R}$, S^1 is closed in $\mathbb{R}^2$. Moreover:

$$\begin{aligned} f_2^{-1}((-\infty, 1]) &= \{(x, y) \mid f((x, y)) \in (-\infty, 1]\} \\ &= \{(x, y) \mid x^2 + y^2 \leq 1\} \\ &= B^2. \end{aligned}$$

Since $(-\infty, 1]$ is closed in $\mathbb{R}$, B^2 is closed in $\mathbb{R}^2$. □